

IMPROVING CHESS PERFORMANCE WITH COGNITIVE-ENHANCING DRUGS

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MOTIVATION

- **When I (Kevin) was growing up, I would play in chess tournaments and often see middle and high schoolers taking caffeine pills before games.**
- **The professional chess governing body has a list of banned substances which includes certain “cognitive-enhancing drugs” and other stimulants including Methamphetamine, but not caffeine.**
- **There is one main study that is cited (see literature slide) but more research into this subject can help players understand the effects.**

- **Do cognitive-enhancing drugs affect chess performance?**
- **Does gender affect chess performance?**
- **Is there an interaction between treatment and sex?**
- **Which treatments are most useful? Are any treatments statistically significant?**



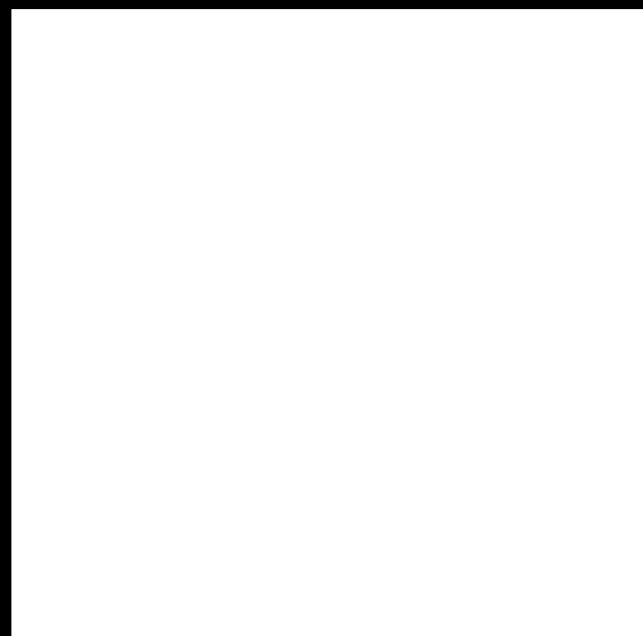
LITERATURE

- **Franke et al., 2017 is a famous study that found an effect of cognitive-enhancing drugs on chess ability, but not on in match performance due to issues playing quickly enough. Caffeine did not have an significant effect on performance in this study.**
- **“Compounds such as methylphenidate and modafinil are used by students in pursuit of better grades, military personnel who need to remain awake for long missions, elderly individuals afraid of cognitive decline and even university academics keen to maintain their performance” - Masud Husain and Mitul A. Mehta on a study about cognitive enhancements by drugs**
- **“Caffeine is known to increase perception of alertness and can induce anxiety, especially at higher doses.” - Simon Cappelletti, Picaentino Daria, Gabriele Sani, and Mariasaria Aromatario on caffeine effects in cognitive areas of the brain**

DESIGN

- **Two-Factor Repeated Measures Design:**
 - **Factor 1: Treatment (Control, Single Caffeine Tablet [100mg], Double Caffeine Tablet [200mg], Methamphetamine Injection [10mg])**
 - **Factor 2: Sex (Male, Female)**
 - **Response: Chess Score over 6 games (Loss = 0, Draw = 0.5, Win = 1)**
 - **Interaction between Treatment and Sex considered**
 - **All players take all treatments, 816 total chess games.**

df = 1



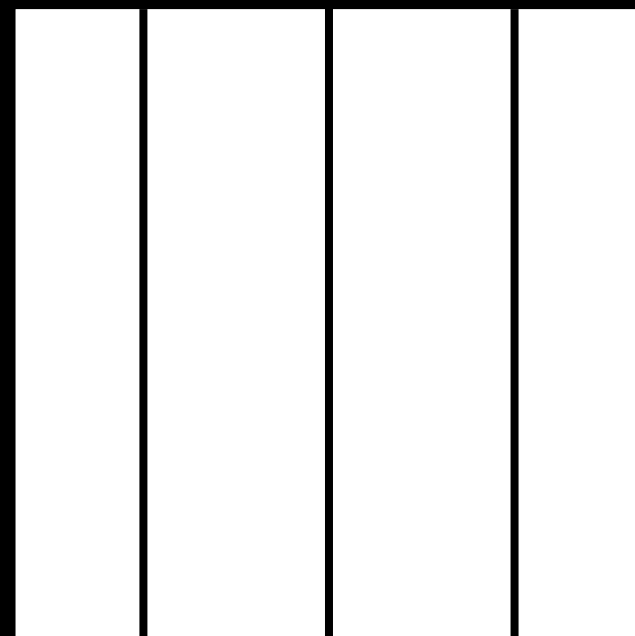
Benchmark

df = 1



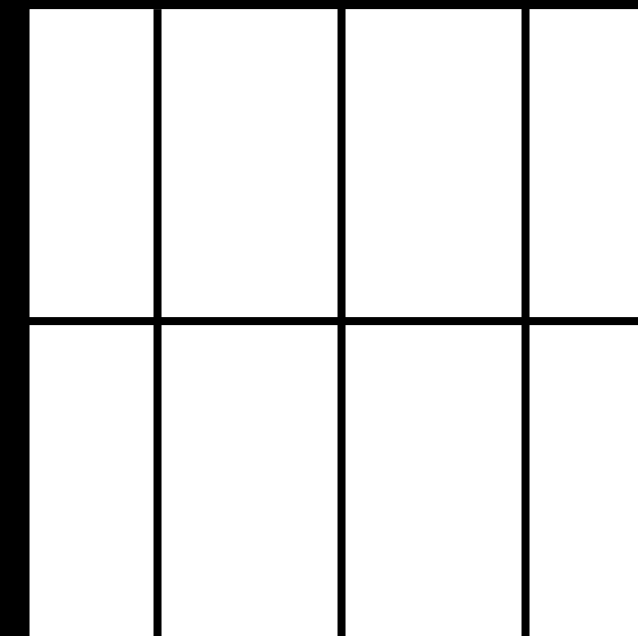
Sex

df = 3



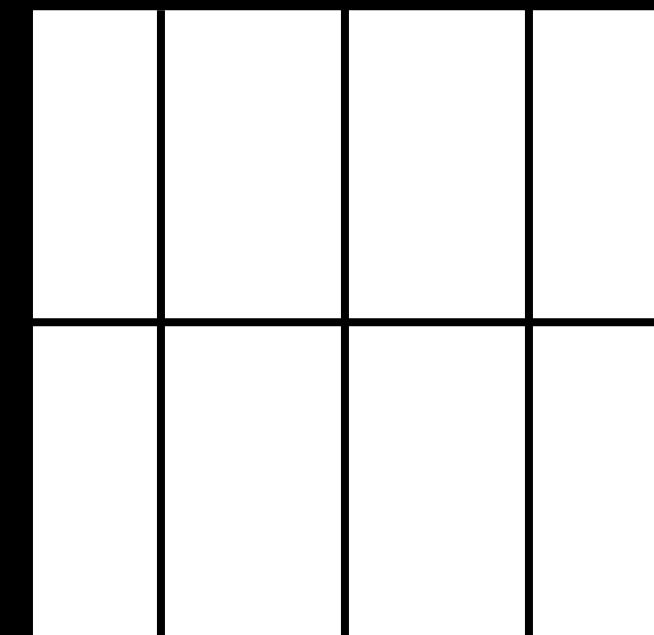
Treatment

df = 3



Interaction

df = 96

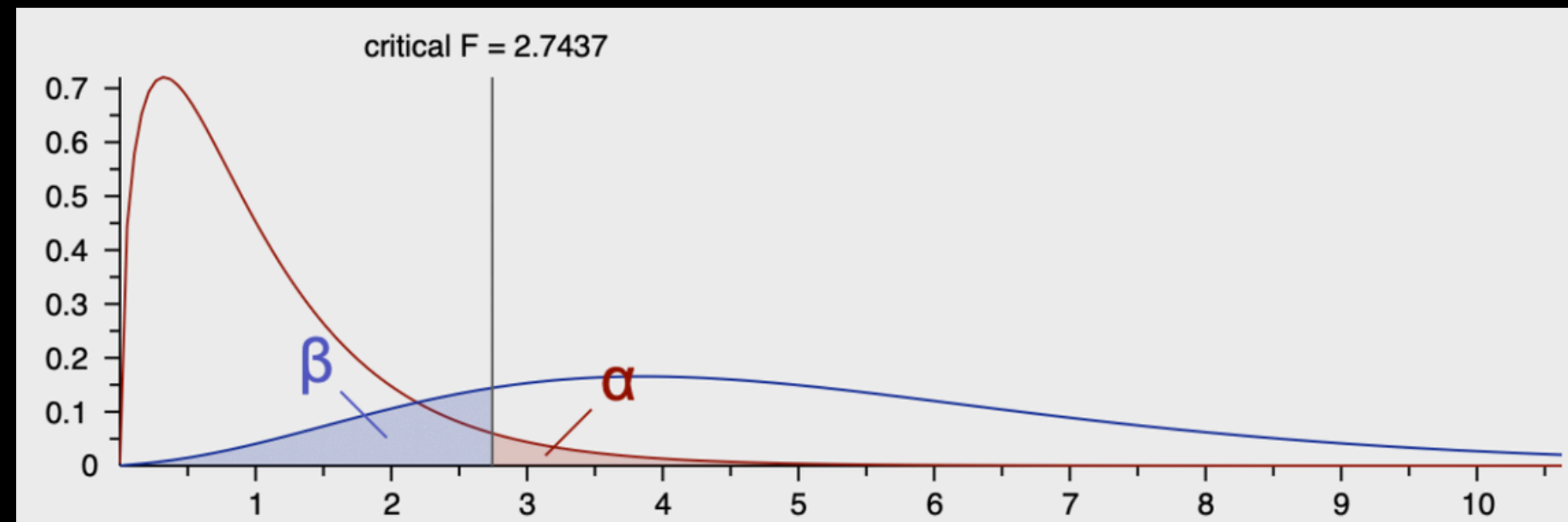


Residuals

SAMPLE CHARACTERISTICS

- **Sample size using Gpower**

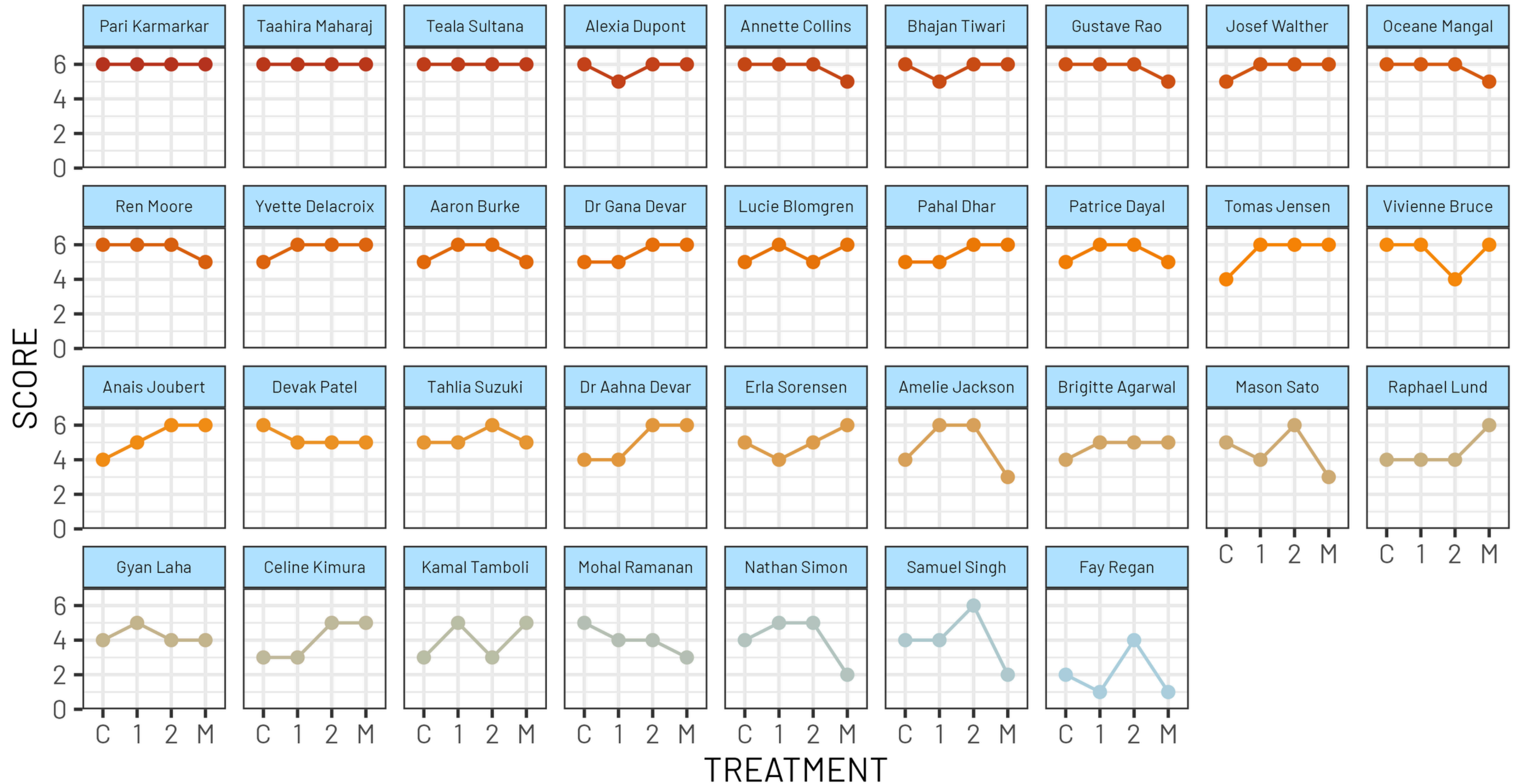
- **Power of 0.8**
- **Effect size of 0.25**
- **Alpha of 0.05**
- **Total sample size necessary determined to be 24**
- **Used $n = 34$, power of 0.94**



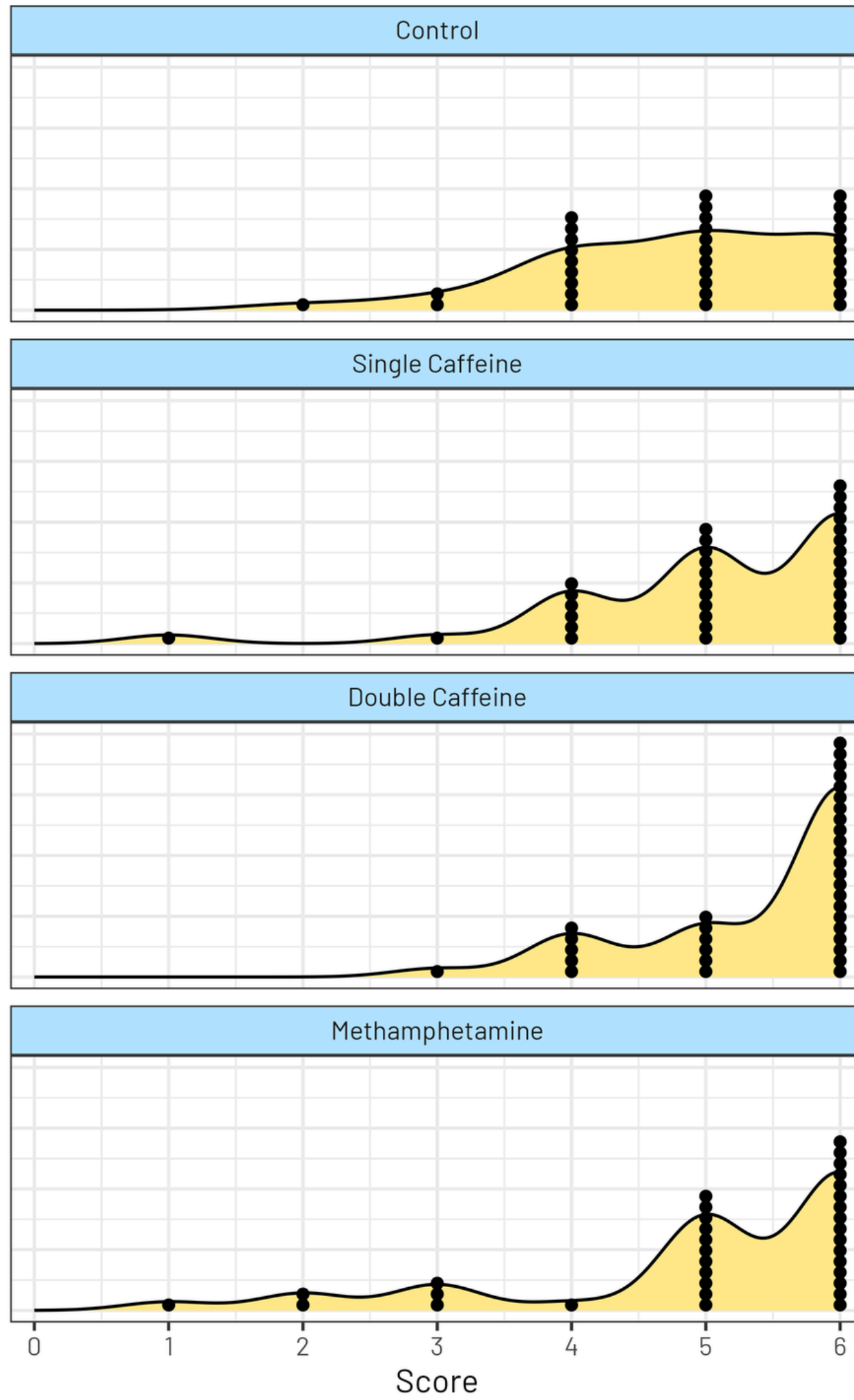
- **Population: Ages 21-29 on Bonne Sante Island, one of the “The Islands”**
- **Randomly selected houses (not cities then houses due to uneven populations)**
- **Sample made up of first 34 eligible and consenting participants**
- **Received a roughly equal male/female breakdown**

Individual Scores

C : Control, 1 : Single Caffeine, 2 : Double Caffeine, M : Methamphetamine

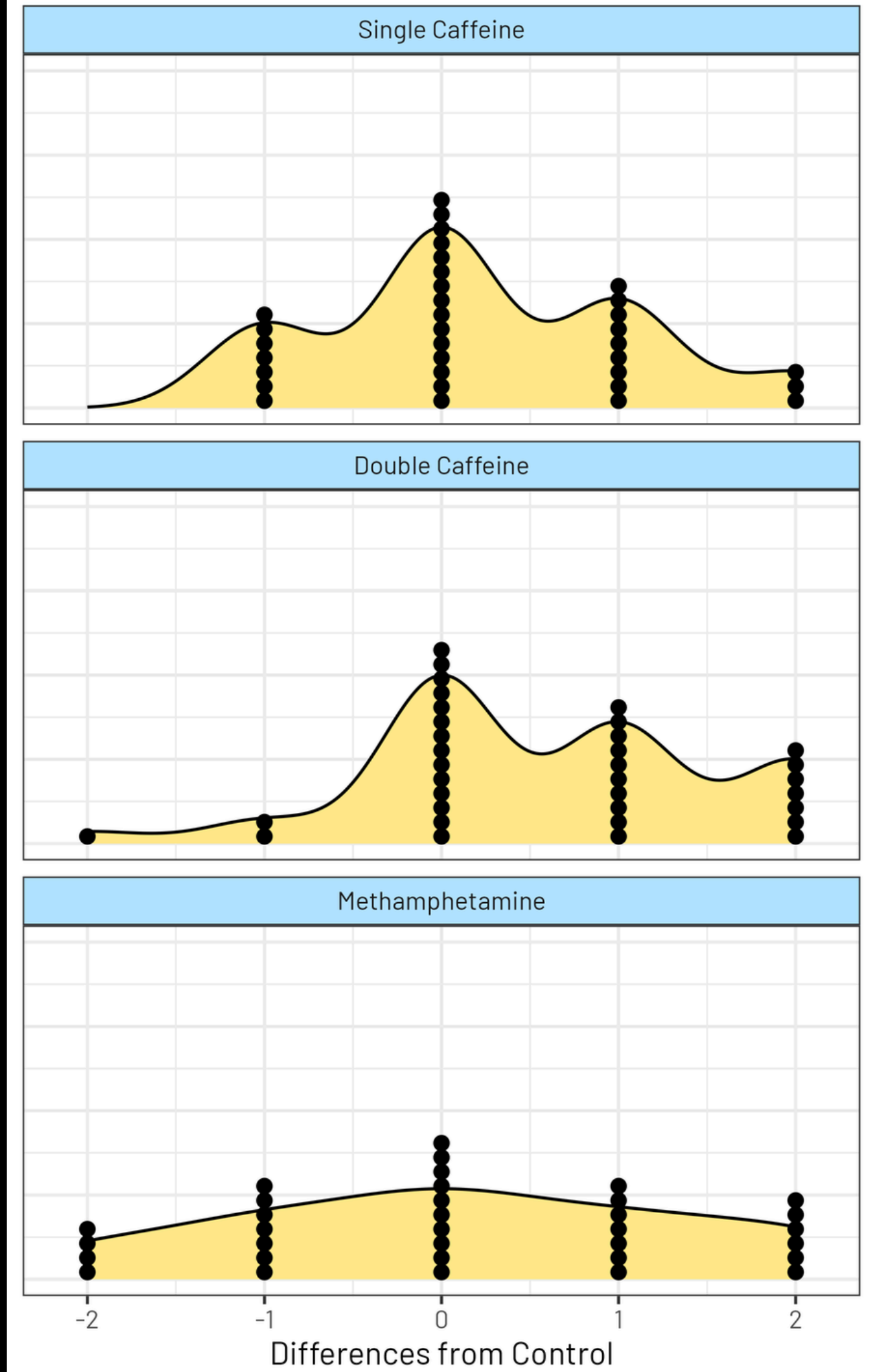


Distribution of Scores



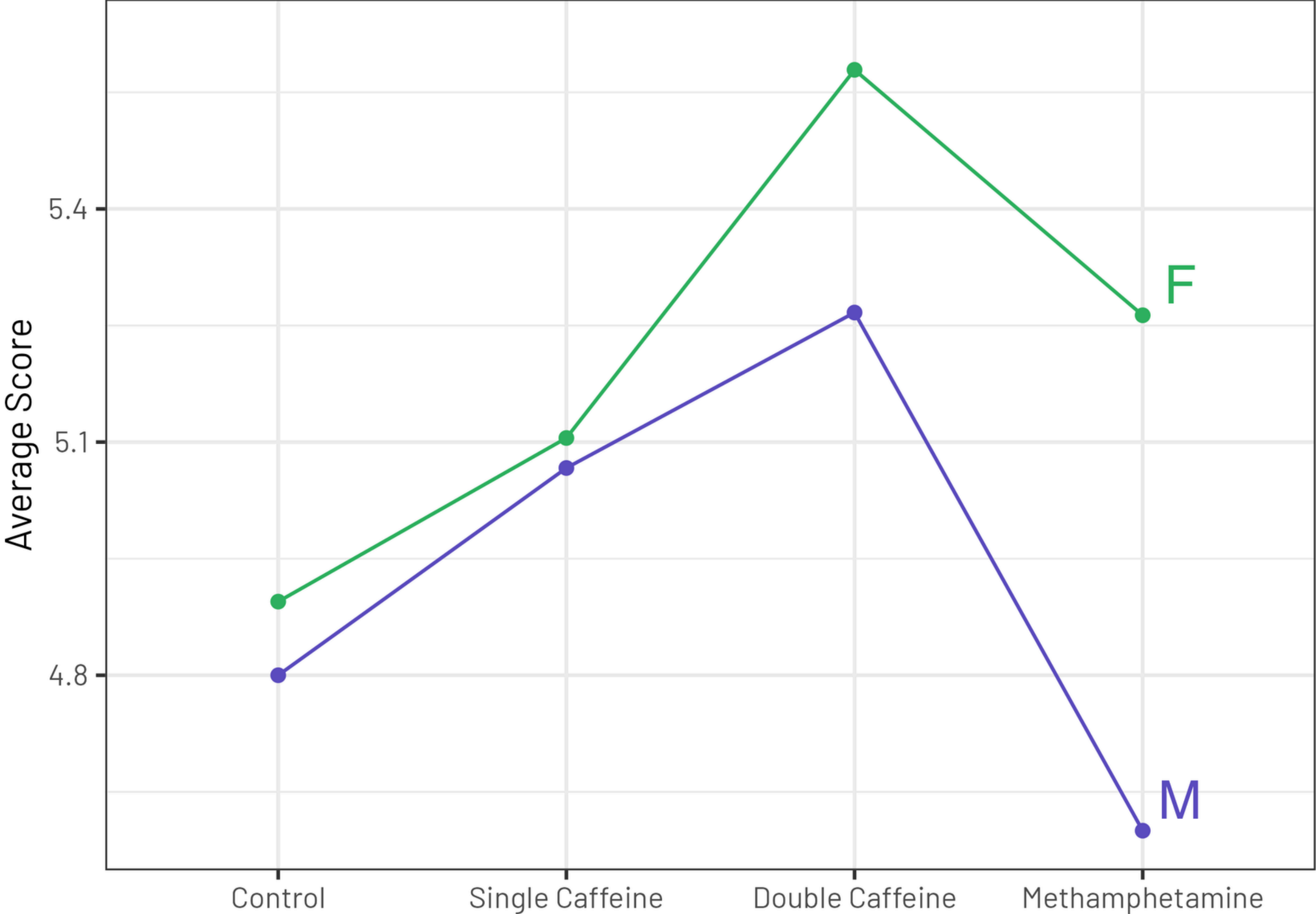
DISTRIBUTIONS

Distribution of Paired Differences



INTERACTION
PLOT

Interaction Plot - Sex vs. Treatment



ANOVA

TREATMENT has p-value < 0.05, can reject the null
and say that not all treatments are equal to the control

stratum	term	df	sumsq	meansq	statistic	p.value
NAME	SEX	1	2.58	2.576	0.869	0.3582
NAME	Residuals	32	94.86	2.965		
Within	TREATMENT	3	6.59	2.196	3.249	0.0252
Within	TREATMENT:SEX	3	2.02	0.672	0.994	0.3992
Within	Residuals	96	64.9	0.676		

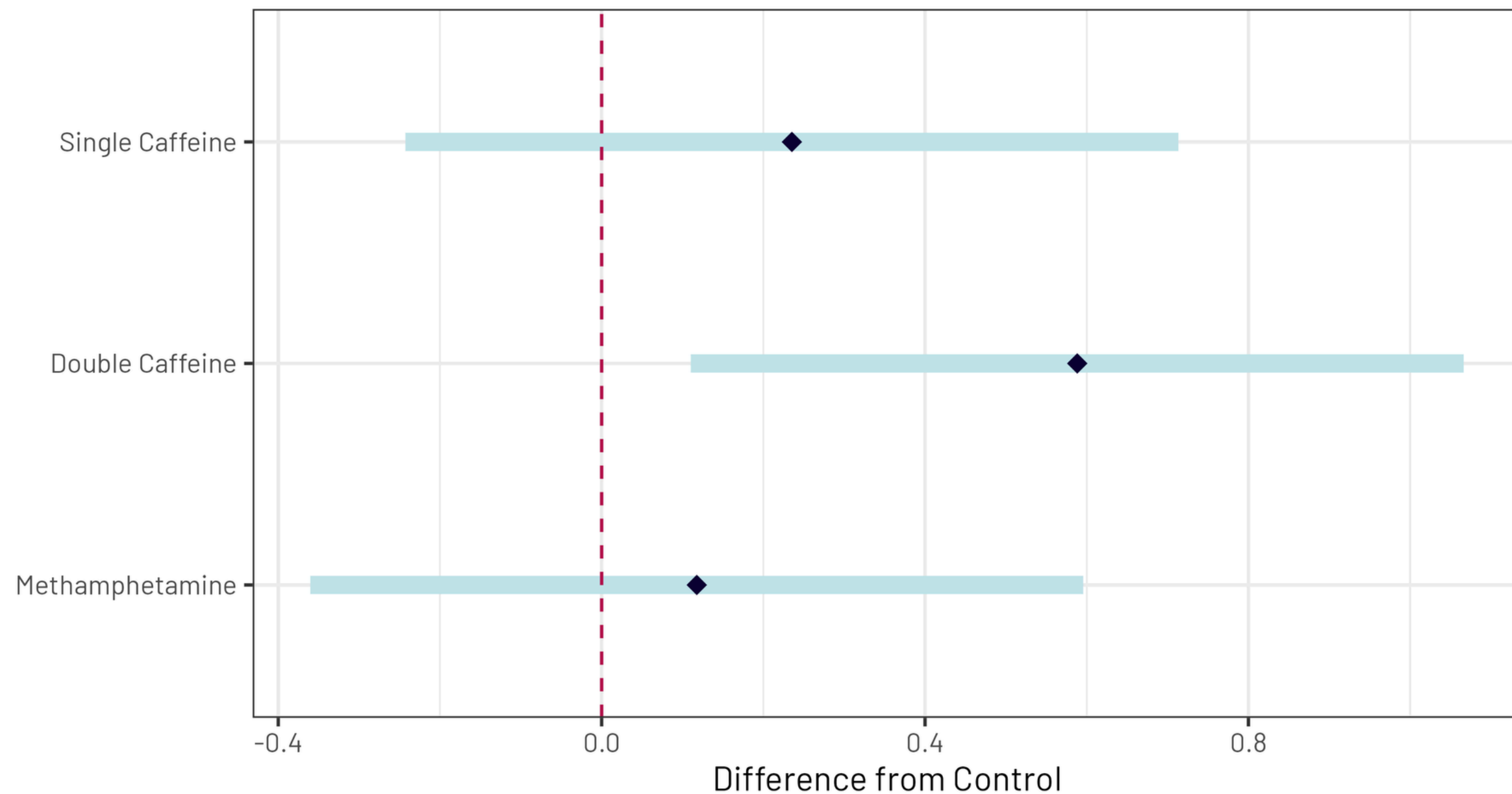
contrast	null.value	estimate	std.error	df	statistic	adj.p.value
Single Caffeine - Control	0	0.235	0.199	99	1.18	0.6408
Double Caffeine - Control	0	0.588	0.199	99	2.95	0.0204
Methamphetamine - Control	0	0.118	0.199	99	0.59	0.9349

POST-
HOC/TUKEYHSD

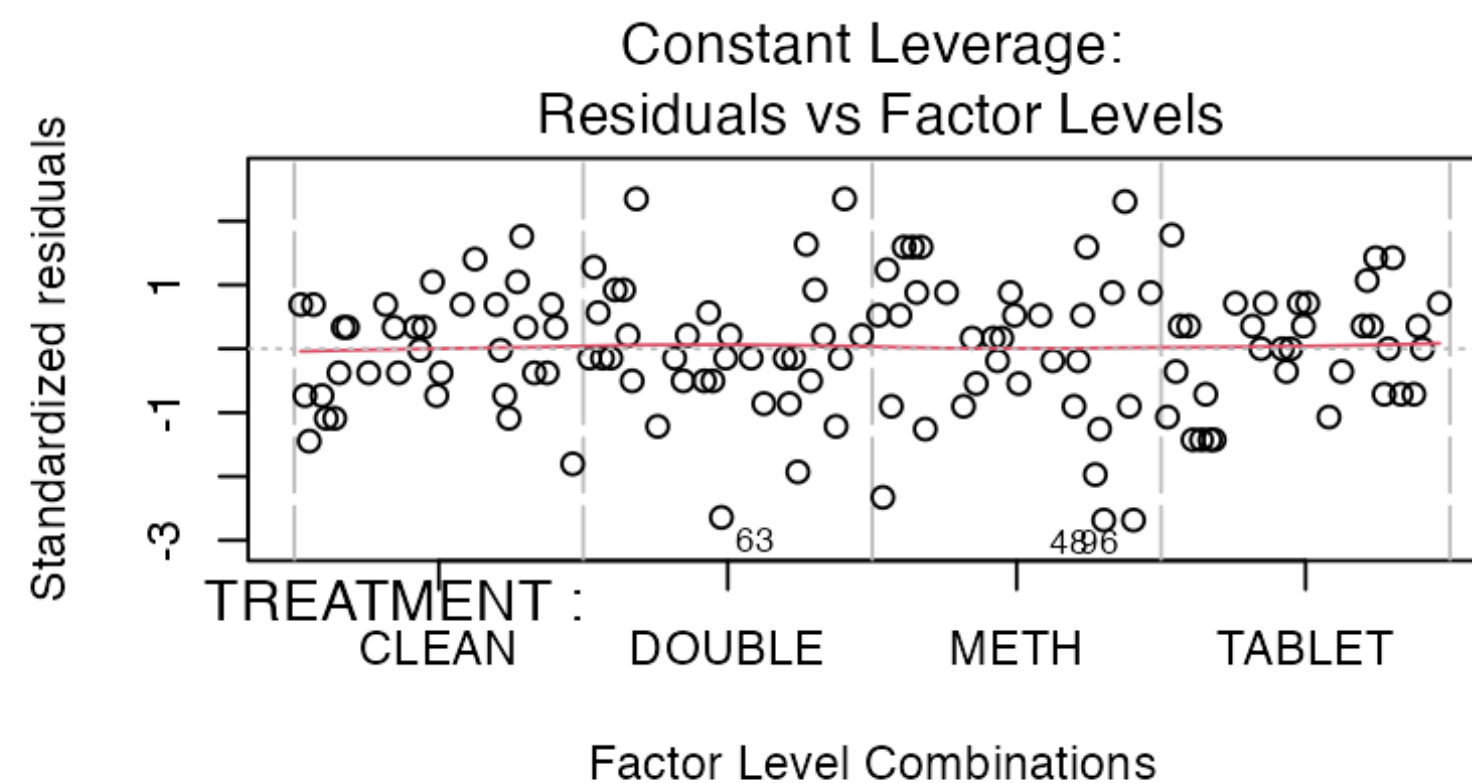
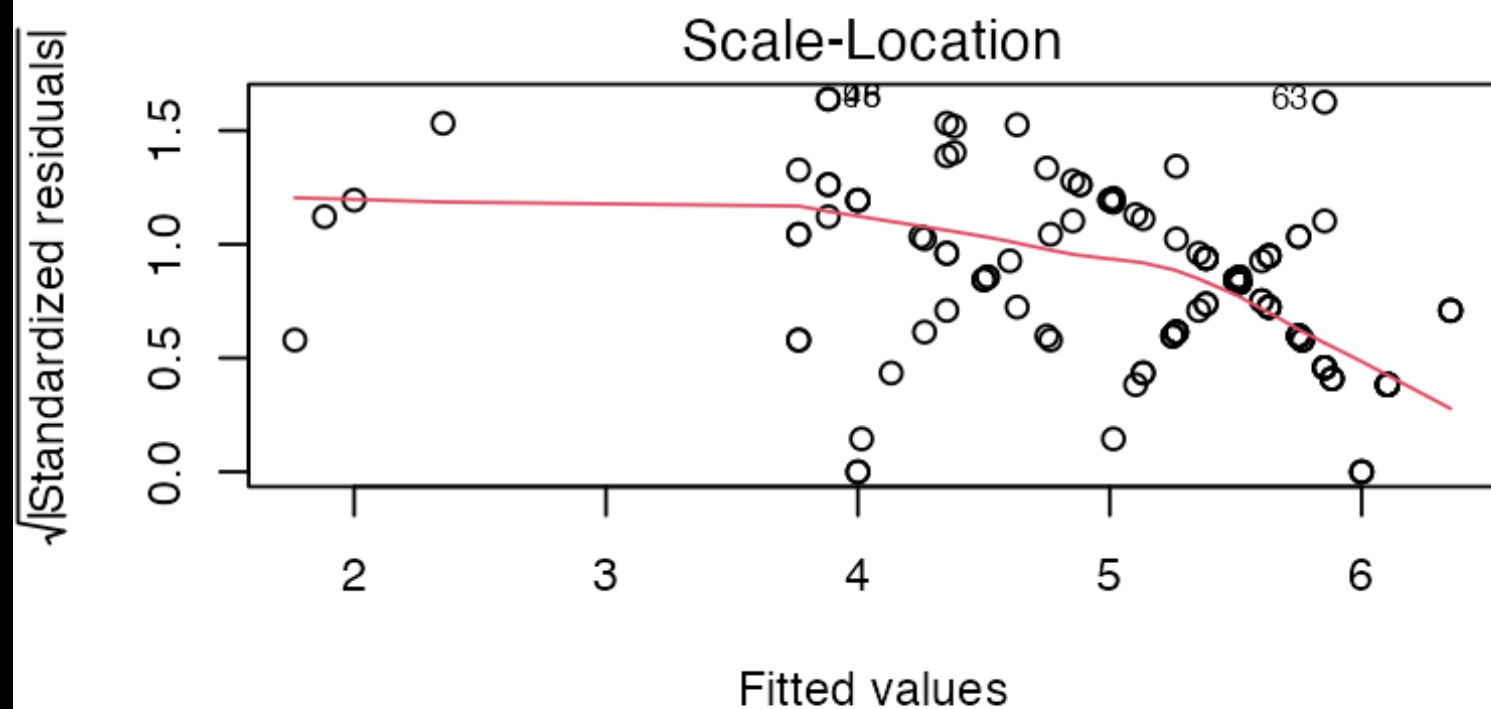
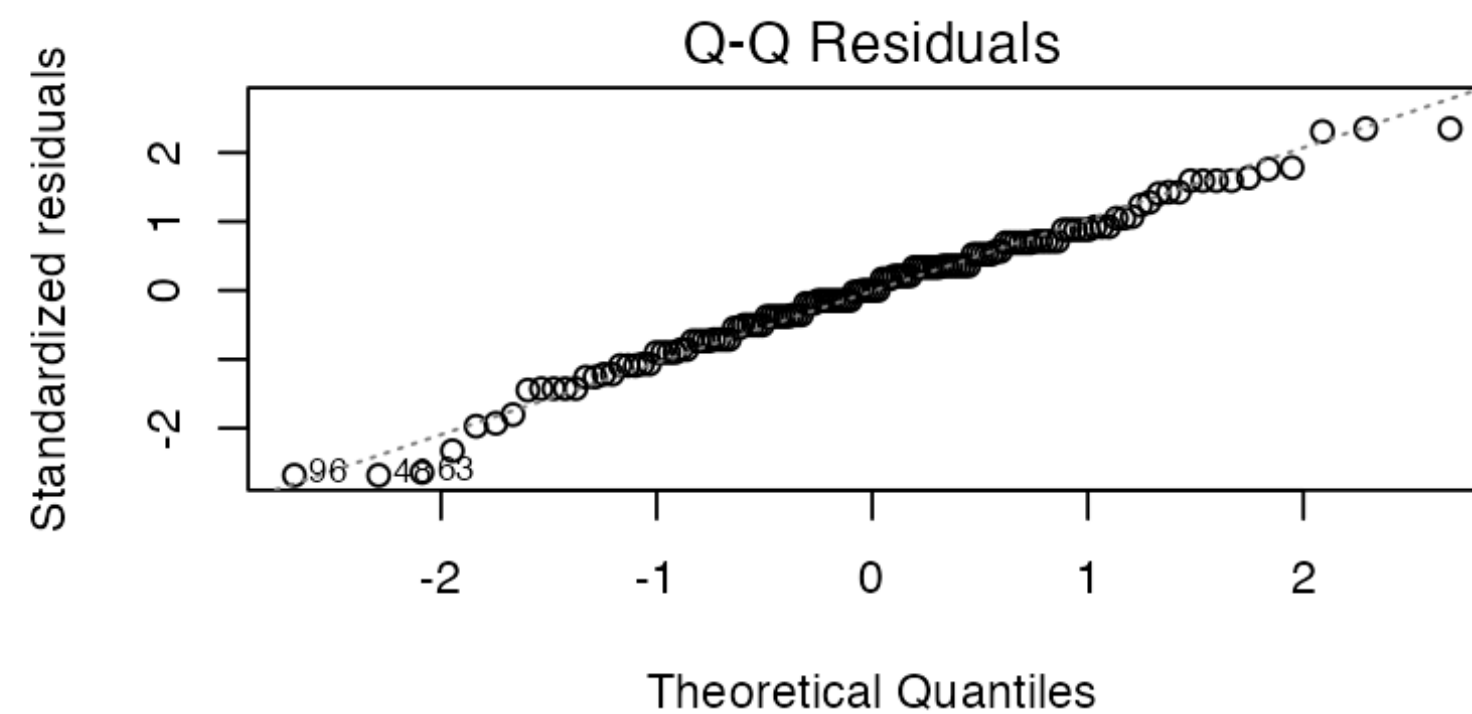
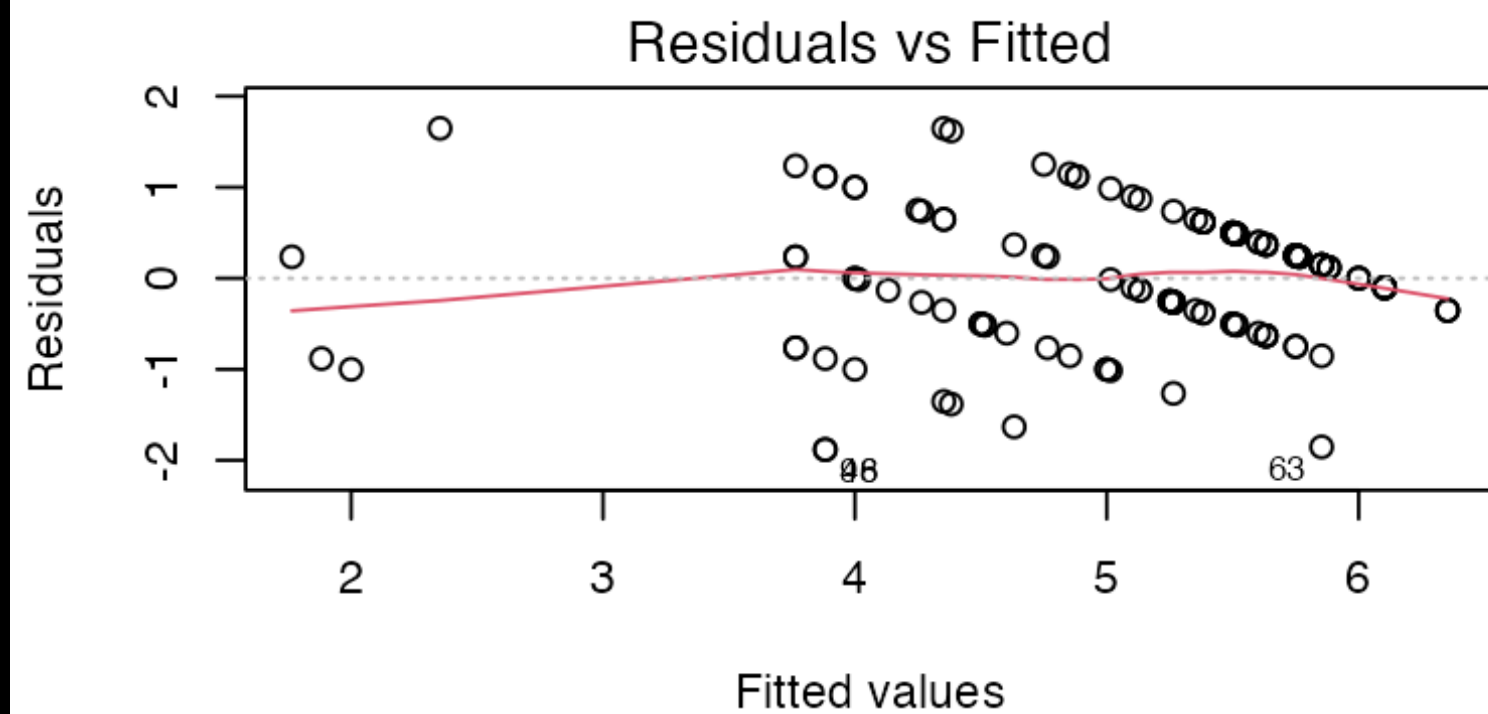
Post-Hoc shows that “Double Caffeine” treatment level is significantly different from “Control”.

Therefore, our study says Double Caffeine improves chess performance in this population.

Tukey Estimates for Difference from Control
95% Confidence Interval



RESIDUALS



CONCLUSION

- **Based on our ANOVA table, with an alpha of 0.05, the choice of treatment is significant to chess performance.**
- **Using our post-hoc analysis, our study shows that taking two caffeine tablets has a positive impact on chess performance outcomes among 21-29 year olds on Bonne Sante Island.**
- **The control, single tablet of caffeine, and methamphetamine outcomes were not significantly different from each other.**

FUTURE RESEARCH QUESTIONS

- **Do cognitive-enhancing drugs help with other tasks like the SAT or Sudoku?**
- **Does caffeine affect the amount of turns it takes to complete a chess game?**
- **How long does it take for the caffeine effect to wear off?**
- **Does this performance benefit stay the same or change for different age groups?**

REFERENCES (1)

- Cappelletti, S. et al. – Caffeine: Cognitive and Physical Performance Enhancer or Psychoactive Drug? – <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4462044/>
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- **Mihailov, E., & Savulescu, J. - Social Policy and Cognitive Enhancement: Lessons from Chess. - Neuroethics 11, 115–127 (2018) - <https://doi.org/10.1007/s12152-018-9354-y>**
- **Bisagno, V., Gonzalez, B., & Urbano, F. - Cognitive Enhancers versus Addictive Psychostimulants: The Good and Bad Side of Dopamine on Prefrontal Cortical Circuits - Pharmacological Research 109, 108-118 (2016) - <https://doi.org/10.1016/j.phrs.2016.01.013>**
- **Caviola L., & Faber, N. - Pills or Push-Ups? Effectiveness and Public Perception of Pharmacological and Non-Pharmacological Cognitive Enhancement - Frontiers in Psychology 6 (2015) - <https://doi.org/10.3389/fpsyg.2015.01852>**